# Math 110 — Week 5

## Quotient maps, isomorphism theorems, dual spaces (Notes Ch. 9–10, LADR 3E–3F, HW5)

## §1 Facts used from Weeks 1–4

(W1) In a finite-dimensional space an independent list extends to a basis; $n$ independent vectors, or $n$ spanning vectors, in a space of dimension $n$ form a basis; a subspace $U \subseteq V$ with $\dim U = \dim V < \infty$ equals $V$. A linear map can be defined by choosing its values on a basis, and two linear maps that agree on a basis are equal (Weeks 2, 3).

(W2) Rank–nullity: $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$. $T$ is injective $\iff \operatorname{null} T = \{0\}$. The image $T(U)$ of a subspace $U$ is a subspace (Week 3).

(W3) If $\dim V = \dim W < \infty$, then injective $\iff$ surjective $\iff$ isomorphism. Finite-dimensional spaces are isomorphic exactly when their dimensions are equal. An isomorphism and its inverse send bases to bases (Week 4).

(W4) $\dim \mathcal{L}(V, W) = \dim V \cdot \dim W$. The matrix of $T$: column $j$ holds the coordinates of $Tv_j$. $\mathcal{M}(ST) = \mathcal{M}(S)\mathcal{M}(T)$ (Week 3).

(W5) Quotient space $V/U$: its elements are the sets $v+U$; $v+U = w+U \iff v-w \in U$; $(v+U)+(w+U) = (v+w)+U$, $\lambda(v+U) = \lambda v + U$. A formula that uses the name $v$ must be checked to be *well-defined*: renaming the same set must not change the answer (Week 4).

(W6) A stochastic matrix $P$ has entries $\geq 0$ and columns summing to 1; $P_{ij}$ is the probability of moving from state $j$ to state $i$, and a distribution $p$ becomes $Pp$ after one step (Week 3).

(W7) If $p(a) = 0$ for a polynomial $p$, then $p = (x-a)q$ for a polynomial $q$ (Week 4, W7).

## §2 This week

### §2.1 The quotient map and the dimension of a quotient (Ch. 9.1)

**The question.** In $V = \mathbb{R}^3$ let $U$ be the line through $(1,1,1)$. The sets $v + U$ are the lines parallel to $U$. How many numbers are needed to say which line a vector lies on, and how do you pick a basis of $V/U$?

**Definition** (Def. 25 / LADR 3.104)**.** For a subspace $U \subseteq V$, the **quotient map** is $\pi\colon V \to V/U$, $\pi(v) = v + U$.

- $\pi$ is linear: by the definition of the operations (W5),

$$\pi(\lambda v + w) = (\lambda v + w) + U = \lambda(v+U) + (w+U) = \lambda\pi(v) + \pi(w).$$

- $\pi$ is surjective: every element of $V/U$ is $v + U = \pi(v)$ for some $v$.
- $\operatorname{null} \pi = U$: $\pi(v)$ is the zero element $0 + U$ exactly when $v - 0 \in U$ (W5).

**Proposition** (Prop. 40 / LADR 3.105: dimension of a quotient)
If $V$ is finite-dimensional, $\dim(V/U) = \dim V - \dim U$.

*Proof.* 1. The quotient map $\pi\colon V \to V/U$ is linear, surjective, and $\operatorname{null}\pi = U$ (bullets above).

2. Rank–nullity (W2) for $\pi$: $\dim V = \dim\operatorname{null}\pi + \dim\operatorname{range}\pi = \dim U + \dim(V/U)$.

□

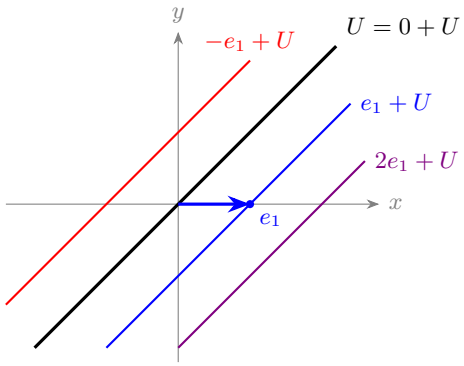
**Proposition** (A basis of $V/U$)
Let $V$ be finite-dimensional, $u_1, \dots, u_k$ a basis of $U$, and $u_1, \dots, u_k, x_1, \dots, x_r$ a basis of $V$. Then $x_1+U, \dots, x_r+U$ is a basis of $V/U$.

*Proof.* 1. $\pi$ is surjective, so it sends the basis $u_1, \dots, u_k, x_1, \dots, x_r$ of $V$ to a spanning list of $V/U$: every $v + U$ is $\pi(\sum a_i u_i + \sum b_j x_j) = \sum a_i \pi(u_i) + \sum b_j \pi(x_j)$.

2. $\pi(u_i) = 0$, so dropping them leaves the spanning list $x_1 + U, \dots, x_r + U$.

3. It has $r = \dim V - \dim U = \dim(V/U)$ elements (Prop. 40), and a spanning list of length dim is a basis (W1).

□




$V = \mathbb{R}^2$, $U = \operatorname{span}((1,1))$. Each line parallel to $U$ is one element of $V/U$, and these elements are exactly the multiples $c\,(e_1 + U) = ce_1 + U$. So $e_1 + U$ is a basis of $V/U$: here $U$ has basis $(1,1)$, which extends to the basis $(1,1), e_1$ of $\mathbb{R}^2$.

**The question answered.** $(1,1,1), e_2, e_3$ is a basis of $\mathbb{R}^3$, so $e_2 + U$, $e_3 + U$ is a basis of $\mathbb{R}^3/U$ and $\dim(\mathbb{R}^3/U) = 3 - 1 = 2$. Two numbers say which line a vector lies on: $(x,y,z) = x(1,1,1) + (y-x)e_2 + (z-x)e_3$, so $(x,y,z) + U = (y-x)(e_2 + U) + (z-x)(e_3 + U)$.

### Shortcuts

- **Basis of $V/U$.** Extend a basis of $U$ to a basis of $V$; the added vectors $+U$ are a basis, and $\dim V/U = \dim V - \dim U$. Reason: the two propositions above. Condition: $V$ finite-dimensional. Instance: for $U = \operatorname{span}((1,1,0)) \subseteq \mathbb{R}^3$, the list $(1,1,0), e_2, e_3$ is a basis of $\mathbb{R}^3$, so $e_2 + U$, $e_3 + U$ is a basis of $\mathbb{R}^3/U$.

### Traps

- Elements of $V/U$ are sets. In $\mathbb{R}^2/\operatorname{span}(e_2)$, $e_1 + U$ and $(e_1 + 5e_2) + U$ are the same element, so "the basis vector $e_1$" of $V/U$ does not make sense; the basis element is the set $e_1 + U$.

## §2.2 The first isomorphism theorem (Ch. 9.1)

**The question.** $T\colon \mathbb{R}^3 \to \mathbb{R}^2$, $T(x,y,z) = (x+y,\, x+y)$, sends $(1,0,0)$ and $(0,1,0)$ to the same vector and misses $(1,0)$. Can we change the domain and the target so that the same formula becomes a bijection?

**Theorem** (Prop. 41 / LADR 3.107: first isomorphism theorem)
Let $T \in \mathcal{L}(V,W)$. The formula

$$\widetilde{T}\colon V/\operatorname{null} T \to \operatorname{range} T, \qquad \widetilde{T}(v + \operatorname{null} T) = Tv$$

defines an isomorphism. Moreover $T = \iota \circ \widetilde{T} \circ \pi$, where $\pi\colon V \to V/\operatorname{null} T$ is the quotient map and $\iota\colon \operatorname{range} T \to W$, $\iota(w) = w$, is the inclusion.

**The idea**

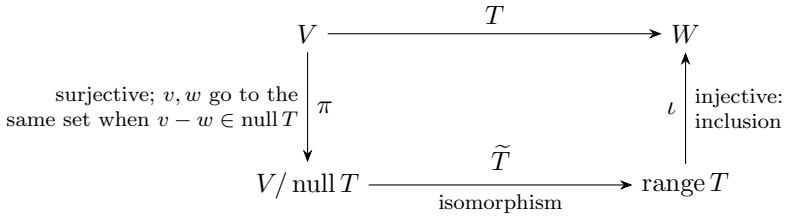
**Surjectivity.** Replacing the target $W$ by $\operatorname{range} T$ makes the map surjective.

**Injectivity.** Passing to the quotient by $\operatorname{null} T$ makes the map injective. Two vectors have the same image exactly when their difference is in $\operatorname{null} T$:

$$Tv = Tw \iff T(v-w) = 0 \iff v - w \in \operatorname{null} T,$$

and $v - w \in \operatorname{null} T$ is exactly the condition for $v$ and $w$ to give the same element of $V/\operatorname{null} T$ (W5). So the pairs of vectors with the same image under $T$ are exactly the pairs that give the same element of $V/\operatorname{null} T$, and on $V/\operatorname{null} T$ different elements have different images.

Read left to right, the chain says the formula $\widetilde{T}(v + \operatorname{null} T) = Tv$ is well-defined (same set $\Rightarrow$ same value). Read right to left it says $\widetilde{T}$ is injective (same value $\Rightarrow$ same set).




Every linear map is a quotient map (surjective), then an isomorphism, then an inclusion (injective). Going down, across and up gives the same result as $T$.

*Proof.*

1. **Well-defined** (the chain read left to right). Suppose $v + \operatorname{null} T = w + \operatorname{null} T$. Then $v - w \in \operatorname{null} T$ (W5), so $Tv - Tw = T(v-w) = 0$, so $Tv = Tw$. The value $\widetilde{T}(v + \operatorname{null} T)$ does not depend on the name $v$.
2. **Linear.** For $v, w \in V$ and $\lambda \in \mathbb{F}$,
$$\widetilde{T}\big(\lambda(v+\operatorname{null} T) + (w+\operatorname{null} T)\big) = \widetilde{T}\big((\lambda v + w) + \operatorname{null} T\big) = T(\lambda v + w) = \lambda Tv + Tw = \lambda\widetilde{T}(v+\operatorname{null} T) + \widetilde{T}(w+\operatorname{null} T).$$
3. **Injective** (the chain read right to left). Suppose $\widetilde{T}(v + \operatorname{null} T) = 0$. Then $Tv = 0$, so $v \in \operatorname{null} T$, so $v + \operatorname{null} T = 0 + \operatorname{null} T$ is the zero element of the quotient.
4. **Surjective onto** $\operatorname{range} T$**.** Every element of $\operatorname{range} T$ is $Tv = \widetilde{T}(v + \operatorname{null} T)$ for some $v$.
5. **Factorisation.** For every $v \in V$,
$$\iota(\widetilde{T}(\pi(v))) = \iota(\widetilde{T}(v + \operatorname{null} T)) = \iota(Tv) = Tv.$$

□

Dimensions:

$$\dim V - \dim \operatorname{null} T = \dim(V/\operatorname{null} T) = \dim \operatorname{range} T.$$

This is rank–nullity again (not a new proof, since Prop. 40 used rank–nullity).

**Example 43: differentiation.** $D\colon \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$, $Dp = p'$, is surjective: every polynomial $q = \sum a_k x^k$ is the derivative of $\sum \frac{a_k}{k+1} x^{k+1}$. Its null space is the constants $\mathbb{R}$. The induced isomorphism is

$$\mathcal{P}(\mathbb{R})/\mathbb{R} \to \mathcal{P}(\mathbb{R}), \qquad p + \mathbb{R} \mapsto p'.$$

Its inverse is the indefinite integral $q \mapsto \int q\,dx$, whose value is a polynomial "$+C$", that is, an element of $\mathcal{P}(\mathbb{R})/\mathbb{R}$.

**Infinite-dimensional $V$: the quotient can have either kind of dimension.** In $V = \mathcal{P}(\mathbb{R})$ let $U = \{p : p(0) = 0\}$. The map $p \mapsto p(0)$ is linear, onto $\mathbb{R}$ (the constant $c$ goes to $c$), with null space $U$. By Prop. 41, $V/U \cong \mathbb{R}$ has dimension 1. For $U = \{0\}$ instead, $V/U \cong V$ is infinite-dimensional: the quotient map $v \mapsto v + \{0\}$ is surjective with null space $\{0\}$, so it is an isomorphism.

Remark from the notes: a subspace $U$ comes with an injective map (its inclusion), and every injective map $i\colon U \to V$ identifies $U$ with the subspace $i(U)$. In the same way a quotient $V/U$ comes with a surjective map $\pi$, and every surjective $T\colon V \to W$ identifies $W$ with the quotient $V/\operatorname{null} T$.

**The question answered.** Yes. The null space and range are

$$\operatorname{null} T = \{(x, y, z) : x + y = 0\} = \operatorname{span}((1, -1, 0), (0, 0, 1)), \qquad \operatorname{range} T = \operatorname{span}((1, 1)).$$

By Prop. 41, $(x, y, z) + \operatorname{null} T \mapsto (x + y, x + y)$ is an isomorphism $\mathbb{R}^3/\operatorname{null} T \to \operatorname{span}((1, 1))$; both have dimension 1.

**Shortcuts**

- **Proving "quotient $\cong X$".** Write a surjective linear map from the numerator onto $X$, find its null space, and quote the first isomorphism theorem; well-definedness is then part of the theorem. Reason: Prop. 41. Instance: $\mathbb{R}^3/\operatorname{span}(e_3) \cong \mathbb{R}^2$, since $(x, y, z) \mapsto (x, y)$ is onto $\mathbb{R}^2$ with null space $\operatorname{span}(e_3)$.
- **Basis of $V/\operatorname{null} T$ when $T$ is onto $\mathbb{F}^k$.** Pick $p_1, \dots, p_k$ with $Tp_i = e_i$; then $p_1 + \operatorname{null} T, \dots, p_k + \operatorname{null} T$ is a basis. Reason: $\widetilde{T}$ is an isomorphism sending $p_i + \operatorname{null} T$ to $e_i$, and the inverse of an isomorphism sends a basis to a basis (W3). Instance: HW5 P1, with $p_1 = 1 - x$, $p_2 = x$.

**Traps**

- $\dim V/U = \dim V - \dim U$ needs $V$ finite-dimensional: $\mathcal{P}(\mathbb{R})/\{p : p(0) = 0\}$ has dimension 1, although $\mathcal{P}(\mathbb{R})$ and $\{p : p(0) = 0\}$ are both infinite-dimensional.
- The target of $\widetilde{T}$ is $\operatorname{range} T$, not $W$: in the question, $\mathbb{R}^3/\operatorname{null} T$ has dimension 1 and is not isomorphic to $\mathbb{R}^2$.

## §2.3 Induced maps; second and third isomorphism theorems (Ch. 9.2–9.3)

**The question.** Differentiation $D$ on $\mathcal{P}(\mathbb{R})$ maps the constants to 0. Does $p + \mathbb{R} \mapsto p'$ give a well-defined map on $\mathcal{P}(\mathbb{R})/\mathbb{R}$? Does $p + \mathbb{R} \mapsto p(0)$? In general, when does a formula $v + U_1 \mapsto Tv + U_2$ make sense?

**Proposition** (Prop. 42: when a map passes to quotients)

Let $T\colon V \to W$ be linear, $U_1 \subseteq V$ and $U_2 \subseteq W$ subspaces with $T(U_1) \subseteq U_2$. Then $\widetilde{T}\colon V/U_1 \to W/U_2$, $\widetilde{T}(v + U_1) = Tv + U_2$, is a well-defined linear map.

**The idea**

The formula uses the name $v$, so the only thing to prove is that renaming does not matter. Two names $v, w$ of the same set differ by an element of $U_1$. Their images $Tv, Tw$ then differ by an element of $T(U_1)$. For the outputs to be the same set we need that difference to be in $U_2$. So the hypothesis $T(U_1) \subseteq U_2$ is what is required. It is also necessary: if $T(U_1) \not\subseteq U_2$, pick $u \in U_1$ with $Tu \notin U_2$; then $u$ and $0$ name the same element of $V/U_1$, but $Tu + U_2 \neq 0 + U_2$.

*Proof.* 1. **Well-defined.** Suppose $v + U_1 = w + U_1$. Then $v - w \in U_1$, so $Tv - Tw = T(v - w) \in T(U_1) \subseteq U_2$, so $Tv + U_2 = Tw + U_2$.

2. **Linear.** For $v, w \in V$ and $\lambda \in \mathbb{F}$,

$$\widetilde{T}(\lambda(v + U_1) + (w + U_1)) = T(\lambda v + w) + U_2 = \lambda(Tv + U_2) + (Tw + U_2).$$

□

**Special case $U_2 = \{0\}$ (HW5 P2).** In $W/\{0\}$ each element $w + \{0\}$ is the one-element set $\{w\}$, and the map $w \mapsto w + \{0\}$ is an isomorphism $W \to W/\{0\}$ (it is the quotient map, surjective, with null space $\{0\}$). So if $U \subseteq \operatorname{null} T$, then $T$ induces $\widetilde{T}\colon V/U \to W$, $\widetilde{T}(v + U) = Tv$. The first isomorphism theorem is the case $U = \operatorname{null} T$.

**Theorem** (Prop. 43: second isomorphism theorem)

For subspaces $U, W \subseteq V$ there is a natural isomorphism

$$\varphi\colon U/(U \cap W) \to (U + W)/W, \qquad \varphi\big(u + (U \cap W)\big) = u + W.$$

"Natural" in the notes means: the isomorphism is given by a formula that uses only the subspaces themselves, with no choice of basis or complement.

**The idea**

The plan for every statement "quotient $\cong$ something" is the same: find a surjective linear map from the numerator onto the something, compute its null space, and apply the first isomorphism theorem. Here the numerator is $U$ and the target is $(U + W)/W$, and the map is $u \mapsto u + W$.

In finite dimensions the theorem gives $\dim U - \dim(U \cap W) = \dim(U + W) - \dim W$, which is the formula for $\dim(U + W)$ from Week 2.

*Proof.* 1. Define $S\colon U \to (U + W)/W$ by $S(u) = u + W$. It is linear, because it is the quotient map of $U + W$ restricted to $U$.

2. $S$ **is surjective.** An element of $(U + W)/W$ is $(u + w) + W$ with $u \in U$, $w \in W$. Since $(u + w) - u = w \in W$, this equals $u + W = S(u)$.

3. $\operatorname{null} S = U \cap W$**.** For $u \in U$: $S(u) = 0 + W \iff u \in W \iff u \in U \cap W$.

4. By the first isomorphism theorem, $U/\operatorname{null} S \to \operatorname{range} S$, that is, $U/(U \cap W) \to (U + W)/W$, $u + (U \cap W) \mapsto u + W$, is an isomorphism.

□

**Theorem** (Prop. 44: third isomorphism theorem)
Let $W \subseteq U \subseteq V$ be subspaces. Then $U/W = \{u + W : u \in U\}$ is a subspace of $V/W$, and

$$\varphi\colon (V/W)/(U/W) \to V/U, \qquad \varphi\big((v + W) + U/W\big) = v + U,$$

is a well-defined isomorphism.

The plan is the same as for Prop. 43: find a surjective map from $V/W$ onto $V/U$ and compute its null space. The candidate is $v + W \mapsto v + U$, which replaces the set $v + W$ by the larger set $v + U$.

*Proof.* 1. $U/W$ **is a subspace.** $U/W$ is the image of the subspace $U$ under the quotient map $V \to V/W$, and the image of a subspace under a linear map is a subspace (W2).

2. The identity map of $V$ sends $W$ into $U$. By Prop. 42 it induces the linear map $S\colon V/W \to V/U$, $S(v+W) = v+U$.

3. $S$ **is surjective**: every $v + U$ is $S(v + W)$.

4. $\operatorname{null} S = U/W$: $S(v + W) = 0 + U \iff v \in U \iff v + W \in U/W$. (For the last step: if $v + W = u + W$ with $u \in U$, then $v = u + (v - u) \in U + W = U$.)

5. The first isomorphism theorem gives the isomorphism $(V/W)/(U/W) \to V/U$, $(v+W)+U/W \mapsto S(v+W) = v+U$. □

- Dimension check: $(\dim V - \dim W) - (\dim U - \dim W) = \dim V - \dim U$.
- Use in induction proofs: if $W \neq \{0\}$, then $V/W$ has smaller dimension than $V$.

**The question answered.** $p + \mathbb{R} \mapsto p'$ is well defined, since $D(\mathbb{R}) = \{0\}$; it is Ex. 43. $p + \mathbb{R} \mapsto p(0)$ is not: $x$ and $x + 1$ name the same element of $\mathcal{P}(\mathbb{R})/\mathbb{R}$ but give 0 and 1. In general, $v + U_1 \mapsto Tv + U_2$ makes sense exactly when $T(U_1) \subseteq U_2$ (Prop. 42 and its idea box).

**Shortcuts**

- **Defining a map out of** $V/U$**.** The formula $v + U \mapsto Tv$ is valid exactly when $U \subseteq \operatorname{null} T$. Reason: Prop. 42 with $U_2 = \{0\}$. Instance: $(x, y, z) + \operatorname{span}(e_3) \mapsto x + 2y$ is well defined, since $(x, y, z) \mapsto x + 2y$ is 0 on $e_3$; $(x, y, z) + \operatorname{span}(e_3) \mapsto z$ is not, since $e_3$ and 0 name the same set but give 1 and 0.

**Traps**

- The second isomorphism theorem has $U \cap W$ in the first denominator, not $W$: $U/W$ is not defined unless $W \subseteq U$. In $\mathbb{R}^3$ with $U = \operatorname{span}(e_1, e_2)$, $W = \operatorname{span}(e_2, e_3)$, the theorem says $U/\operatorname{span}(e_2) \cong \mathbb{R}^3/W$; both have dimension 1.

## §2.4 Dual space and dual basis (Ch. 10.1)

**The question.** A vector can be studied through numbers that depend linearly on it: a coordinate, the value of a polynomial at a point, an integral. What is the space of all such functions, and what is a basis of it when $V = \mathbb{R}^2$ with the basis $(1, 1), (1, -1)$?

**Definition** (Def. 26, 27 / LADR 3.108, 3.112)**.** A **linear functional** on $V$ is a linear map $\varphi\colon V \to \mathbb{F}$. The **dual space** is $V' = \mathcal{L}(V, \mathbb{F})$. If $v_1, \ldots, v_n$ is a basis of $V$, its **dual basis** is the list $\varphi_1, \ldots, \varphi_n \in V'$ with

$$\varphi_j(v_k) = \delta_{jk} = \begin{cases} 1 & j = k \\ 0 & j \neq k. \end{cases}$$

- Elements of $V'$ are also called linear forms, one-forms or covectors; these are different names for the same objects.

- $V'$ has nothing to do with derivatives. Other books (and the practice midterm) write $V^*$. The notes later use the asterisk for a different operation, so in this course write $V'$.
- Examples on $\mathcal{P}(\mathbb{R})$ (LADR 3.109): $p \mapsto p(4)$, $p \mapsto 3p''(5) + 7p(4)$, $p \mapsto \int_0^1 p(x)\,dx$. Each is linear because evaluation, differentiation and integration are.
- $\varphi_j$ exists and is unique because a linear map is determined by its values on a basis (W1).
- **What $\varphi_j$ does:** $\varphi_j(a_1v_1 + \cdots + a_nv_n) = a_j$. It reads off the $j$-th coordinate.
- Ex. 44: for $\mathbb{F}^n$ (columns) a functional is a row vector, acting by row times column. The dual basis of $e_1, \ldots, e_n$ is the list of rows $(1\ 0\ \cdots\ 0), \ldots, (0\ \cdots\ 0\ 1)$.
- On $\mathbb{R}^n$ the dot product gives a map $v \mapsto (w \mapsto w \cdot v)$, which sends a column $v$ to the row $v^{\mathsf{T}}$. It is an isomorphism $\mathbb{R}^n \to (\mathbb{R}^n)'$, but it depends on the dot product: a different inner product gives a different map. The double-dual map in the next section uses nothing of this kind.
- $\dim V' = \dim V \cdot \dim \mathbb{F} = \dim V$ (W4). So $V' \cong V$ when $V$ is finite-dimensional. But an isomorphism has to be chosen, for example $v_j \mapsto \varphi_j$, and a different basis gives a different one. For $V = \mathbb{R}^2$: the basis $e_1, e_2$ gives $e_1 \mapsto (1\ 0)$; the basis $e_1, (1,1)$ has the dual basis $(1\ -1), (0\ 1)$, so it gives $e_1 \mapsto (1\ -1)$. The two isomorphisms send $e_1$ to different rows.

**Proposition** (Prop. 45 / LADR 3.116: the dual basis is a basis)
If $v_1, \ldots, v_n$ is a basis of $V$, its dual basis $\varphi_1, \ldots, \varphi_n$ is a basis of $V'$.

**The idea**

The technique for functionals: *to get information out of an equation between functionals, evaluate both sides on basis vectors.* Evaluating $\lambda_1\varphi_1 + \cdots + \lambda_n\varphi_n$ at $v_k$ makes every term vanish except the $k$-th, and returns $\lambda_k$.

*Proof.* 1. Suppose $\lambda_1\varphi_1 + \cdots + \lambda_n\varphi_n = 0$ (the zero functional). Evaluate at $v_k$:

$$0 = \lambda_1\varphi_1(v_k) + \cdots + \lambda_n\varphi_n(v_k) = \lambda_k.$$

This holds for every $k$, so $\lambda_1 = \cdots = \lambda_n = 0$: the list is independent.

2. It has $n = \dim V'$ vectors, so it is a basis (W1).

□

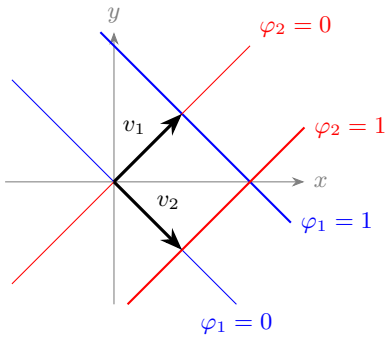
**Coordinates.** Of a vector: $v = \sum_j \varphi_j(v)\, v_j$. Of a functional: $\psi = \sum_j \psi(v_j)\, \varphi_j$ (HW5 P4). The coefficient of $\varphi_j$ is found by evaluating $\psi$ at $v_j$.

**Dual basis of $(1,1), (1,-1)$ in $\mathbb{R}^2$.**

1. Put the basis vectors as the columns of $M$. Row $j$ of $M^{-1}$ times column $k$ of $M$ is entry $(j,k)$ of $M^{-1}M = I$, which is $\delta_{jk}$. So the rows of $M^{-1}$ are the dual basis.

$$M = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \qquad M^{-1} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{bmatrix}.$$

2. So $\varphi_1(x,y) = \frac{1}{2}(x+y)$ and $\varphi_2(x,y) = \frac{1}{2}(x-y)$. Check: $\varphi_1(1,1) = 1$, $\varphi_1(1,-1) = 0$, $\varphi_2(1,1) = 0$, $\varphi_2(1,-1) = 1$.
3. For $\psi(x,y) = 3x + y$: $\psi = \psi(1,1)\varphi_1 + \psi(1,-1)\varphi_2 = 4\varphi_1 + 2\varphi_2$.




Level lines of the dual basis of $v_1 = (1, 1)$, $v_2 = (1, -1)$. The tip of $v_1$ lies on $\varphi_1 = 1$ and $\varphi_2 = 0$; the tip of $v_2$ lies on $\varphi_1 = 0$ and $\varphi_2 = 1$.

**The question answered.** The space is $V'$, and for $\mathbb{R}^2$ with the basis $(1, 1), (1, -1)$ a basis of it is the dual basis $\varphi_1 = (\frac{1}{2}\ \frac{1}{2})$, $\varphi_2 = (\frac{1}{2}\ -\frac{1}{2})$ of the example.

**Shortcuts**

- **Equation between functionals: evaluate on a basis.** The $j$-th coordinate of $\psi$ in the dual basis is $\psi(v_j)$. Reason: HW5 P4(b). Instance: to show $\psi = 2\varphi_1 - \varphi_2$, check both sides at $v_1$ (value 2) and at $v_2$ (value $-1$).
- **Dual basis in $\mathbb{F}^n$: rows of $M^{-1}$**, where $M$ has the basis vectors as columns. Reason: $M^{-1}M = I$ (the example above). Instance: for the basis $(1, 0), (1, 1)$, $M^{-1} = \left[\begin{smallmatrix} 1 & -1 \\ 0 & 1 \end{smallmatrix}\right]$, so the dual basis is $(1\ -1)$, $(0\ 1)$.
- **A nonzero functional is surjective**, so its null space has dimension $\dim V - 1$. Reason: HW5 P3 and rank–nullity. Condition: $V$ finite-dimensional for the dimension count. Instance: $\varphi(x, y, z) = x + y$ on $\mathbb{R}^3$ has a null space of dimension 2.
- **Known dual bases on $\mathcal{P}_n(\mathbb{R})$.** The dual basis of $1, x, \ldots, x^n$ is $\varphi_k(p) = p^{(k)}(0)/k!$, the $k$-th Taylor coefficient. The functionals $p \mapsto p(c_0), \ldots, p \mapsto p(c_n)$ at distinct points are the dual basis of the Lagrange polynomials $\ell_0, \ldots, \ell_n$ with $\ell_j(c_k) = \delta_{jk}$. Reason: check $\varphi_j(b_k) = \delta_{jk}$ directly. Instance: on $\mathcal{P}_1(\mathbb{R})$ with points $0, 1$, the Lagrange basis is $1 - x$, $x$ (the polynomials of HW5 P1), dual to $p \mapsto p(0)$, $p \mapsto p(1)$.

**Traps**

- The dual basis depends on the whole basis. For $e_1, e_2$, $\varphi_1 = (1\ 0)$. For $e_1, (1, 1)$, $\varphi_1 = (1\ -1)$: changing $v_2$ changed $\varphi_1$.
- $V'$ is the dual space, not a derivative, and $\varphi_j$ is not the $j$-th vector $v_j$ written as a row: for the basis $(1, 1), (1, -1)$, $\varphi_1 = (\frac{1}{2}\ \frac{1}{2})$, not $(1\ 1)$.

## §2.5 The double dual (Ch. 10.1)

**The question.** There is no preferred isomorphism $V \to V'$. Is there one from $V$ to $V'' = (V')'$? For $V = \mathbb{R}^2$, which element of $V''$ should correspond to $(3, 5)$?

**Definition** (Notes 10.1 / LADR 3F: the double dual map)**.** For $v \in V$ let $j_V(v) \in V''$ be "evaluate at $v$": $j_V(v)(\varphi) = \varphi(v)$ for $\varphi \in V'$. This gives $j_V : V \to V''$.

- $j_V(v)$ is linear in $\varphi$: $(\lambda\varphi + \psi)(v) = \lambda\varphi(v) + \psi(v)$. So $j_V(v) \in V''$.
- $j_V$ is linear in $v$: $\varphi(\lambda v + w) = \lambda\varphi(v) + \varphi(w)$.
- The definition uses no basis and no inner product.

**Proposition** (Prop. 46 / LADR 3F: $V \to V''$)

$j_V$ is injective. If $V$ is finite-dimensional, $j_V$ is an isomorphism.

**The idea**

Injective means: if $v \neq 0$ then $j_V(v) \neq 0$, that is, some functional $\varphi$ has $\varphi(v) \neq 0$. So the task is to build a functional that does not vanish at a given nonzero vector. Functionals are built by prescribing values on a basis, so make $v$ a basis vector and give it the value 1.

Restated: if $\varphi(v) = \varphi(w)$ for every $\varphi \in V'$, then $v = w$. If two vectors have the same value under every functional, they are equal.

*Proof.* 1. Let $v \neq 0$. The list $v$ is independent; extend it to a basis $v, v_2, \dots$ of $V$ (W1 if $V$ is finite-dimensional; for infinite-dimensional $V$ we cite that every independent list extends to a basis, a consequence of Zorn's lemma that this course does not prove).

2. Let $\varphi$ be the functional with $\varphi(v) = 1$ and $\varphi(v_k) = 0$ for the other basis vectors.

3. Then $j_V(v)(\varphi) = \varphi(v) = 1 \neq 0$, so $j_V(v) \neq 0$. Hence $\operatorname{null} j_V = \{0\}$.

4. If $\dim V < \infty$: $\dim V'' = \dim V' = \dim V$, and an injective map between spaces of equal finite dimension is an isomorphism (W3).

□

**In coordinates.** Vectors of $\mathbb{F}^n$ are columns and functionals are rows. A functional on rows is determined by its values on the rows $e_1^{\mathsf{T}}, \dots, e_n^{\mathsf{T}}$, that is, by $n$ numbers, which we write as a column. $j_V(v)$ is the column $v$ itself, since $j_V(v)(e_k^{\mathsf{T}}) = e_k^{\mathsf{T}} v = v_k$.

**The question answered.** Yes: $j_V$ is defined without choices and is an isomorphism for finite-dimensional $V$. For $(3,5) \in \mathbb{R}^2$, $j_V(3,5)$ is the function on rows $(a\ b) \mapsto 3a + 5b$; in coordinates it is the column $(3,5)$ again.

**Shortcuts**

- **To show $v = 0$, find a functional with $\varphi(v) \neq 0$**, or check that every element of a dual basis vanishes at $v$. Reason: Prop. 46 and HW5 P4(a). Instance: if $\varphi_1(v) = \varphi_2(v) = 0$ for the dual basis of a basis of $\mathbb{R}^2$, then $v = \varphi_1(v)v_1 + \varphi_2(v)v_2 = 0$.

**Traps**

- $V \cong V'$ in finite dimensions, but not by a preferred map (the example with $e_1$ above); $V \cong V''$ by the preferred map $j_V$. In infinite dimensions $j_V$ is injective but need not be surjective (cited; not proved in this course).

## §2.6 The dual map and the transpose (Ch. 10.2)

**The question.** Let $T\colon \mathbb{R}^2 \to \mathbb{R}$, $T(x,y) = 2x - y$ (Practice Midterm 2), and let $\ell(t) = 3t$ be a functional on $\mathbb{R}$. Then $\ell \circ T$ is a functional on $\mathbb{R}^2$. What is it, and what does the map $\ell \mapsto \ell \circ T$ look like as a matrix?

**Definition** (Def. 28 / LADR 3.118)**.** For $T \in \mathcal{L}(V,W)$ the **dual map** is $T'\colon W' \to V'$, $T'(\varphi) = \varphi \circ T$.

- $T'\varphi = \varphi \circ T$ is a composition of linear maps $V \to W \to \mathbb{F}$, so it is linear: $T'\varphi \in V'$.
- $T'$ is linear: for $\varphi, \psi \in W'$, $\lambda \in \mathbb{F}$ and $v \in V$,

$$(T'(\lambda\varphi + \psi))(v) = (\lambda\varphi + \psi)(Tv) = \lambda\varphi(Tv) + \psi(Tv) = (\lambda T'\varphi + T'\psi)(v).$$

**The idea**

Why does the direction reverse? Given $\varphi \in W'$ and $T\colon V \to W$, we get a functional on $V$: first send $v$ to $W$, then apply $\varphi$. That is $\varphi \circ T$. So $T'$ sends functionals on $W$ to functionals on $V$. There is no way to go the other way: a functional on $V$ gives no values on vectors of $W$ outside the range.

The defining equation to use in every computation: $(T'\varphi)(v) = \varphi(Tv)$.

**Example 45 and Practice Midterm 2. (45)** $T\colon \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$, $Tp = xp$, and $\varphi(p) = p(2)$. Then $(T'\varphi)(p) = \varphi(xp) = 2p(2)$, so $T'\varphi = 2\varphi$.

**(PM2)** $T(x, y) = 2x - y$ from $\mathbb{R}^2$ to $\mathbb{R}$ and $\ell(t) = 3t$. Then $(T'\ell)(x, y) = \ell(2x - y) = 6x - 3y$.

**Dual of differentiation (LADR 3.119).** Let $D\colon \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$, $Dp = p'$. For $\varphi(p) = p(3)$: $(D'\varphi)(p) = \varphi(p') = p'(3)$. For $\psi(p) = \int_0^1 p$:

$$(D'\psi)(p) = \int_0^1 p'(x)\,dx = p(1) - p(0),$$

so $D'\psi$ is the functional $p \mapsto p(1) - p(0)$.

**Proposition** (Prop. 47 / LADR 3.120: algebra of dual maps)
The map $\mathcal{L}(V, W) \to \mathcal{L}(W', V')$, $T \mapsto T'$, is linear: $(\lambda T_1 + T_2)' = \lambda T_1' + T_2'$. It reverses products: $(ST)' = T'S'$ for $T \in \mathcal{L}(U, V)$, $S \in \mathcal{L}(V, W)$.

*Proof.* 1. For $\varphi \in W'$ and $v \in V$:

$$\big((\lambda T_1 + T_2)'\varphi\big)(v) = \varphi(\lambda T_1 v + T_2 v) = \lambda\varphi(T_1 v) + \varphi(T_2 v) = \big(\lambda T_1'\varphi + T_2'\varphi\big)(v).$$

2. Apply the definition of the dual map three times, using that composition is associative:

$$(ST)'\varphi = \varphi \circ (S \circ T) = (\varphi \circ S) \circ T = (S'\varphi) \circ T = T'(S'\varphi).$$

□

**Proposition** (Prop. 48: $j$ is compatible with every linear map)
For $T\colon V \to W$: $T'' \circ j_V = j_W \circ T$.

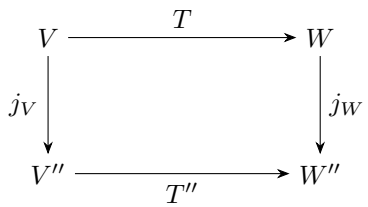
*Proof.* 1. Fix $v \in V$. Two elements of $W''$ are equal when they agree at every $\psi \in W'$.

2. For $\psi \in W'$, unwind the definitions one at a time:

$$\big(T''(j_V(v))\big)(\psi) = j_V(v)(T'\psi) = (T'\psi)(v) = \psi(Tv) = j_W(Tv)(\psi).$$

3. So $T''(j_V(v)) = j_W(Tv)$ for every $v$.

□




Prop. 48: across then down equals down then across.

This is what "canonical" means for $j$. When $V$ and $W$ are finite-dimensional, $j_V$ and $j_W$ are isomorphisms (Prop. 46), and the equation says $T'' = j_W \circ T \circ j_V^{-1}$: after identifying $V$ with $V''$ and $W$ with $W''$, the map $T''$ is $T$.

**Theorem** (Prop. 49 / LADR 3.132: the matrix of the dual map is the transpose)
Let $B = (v_1, \dots, v_n)$, $C = (w_1, \dots, w_m)$ be bases of $V, W$ with dual bases $B' = (v_1', \dots, v_n')$, $C' = (w_1', \dots, w_m')$. If $A = \mathcal{M}(T)$ with respect to $B, C$, then $\mathcal{M}(T')$ with respect to $C', B'$ is $A^{\mathsf{T}}$, where $(A^{\mathsf{T}})_{ij} = a_{ji}$.

**The idea**

Column $j$ of $\mathcal{M}(T')$ must hold the coordinates of the functional $T'(w_j') = w_j' \circ T$ in the basis $v_1', \dots, v_n'$. What does $w_j' \circ T$ do? It sends $v$ to the $j$-th coordinate of $Tv$. In coordinates, the $j$-th coordinate of $Ax$ is $a_{j1}x_1 + \cdots + a_{jn}x_n$: the coefficients are *row $j$ of $A$*. So row $j$ of $A$ becomes column $j$ of $\mathcal{M}(T')$. That is the transpose.

To prove it, find the coordinates of $T'(w_j')$ by evaluating at each $v_k$ (the technique from Prop. 45).

*Proof.* 1. By definition of $A$: $Tv_k = \sum_{l=1}^{m} a_{lk} w_l$.

2. Evaluate $T'(w_j')$ at $v_k$:
$$T'(w_j')(v_k) = w_j'(Tv_k) = w_j'\Big(\sum_l a_{lk} w_l\Big) = a_{jk}.$$

3. The coordinates of a functional $\psi$ in the dual basis are its values $\psi(v_k)$ (HW5 P4(b)). So $T'(w_j') = \sum_{k=1}^{n} a_{jk} v_k'$.

4. So column $j$ of $\mathcal{M}(T')$ is $(a_{j1}, \dots, a_{jn})^{\mathsf{T}}$, which is row $j$ of $A$.

□

**Example 46.** Let
$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} : \mathbb{R}^3 \to \mathbb{R}^2.$$
The dual map sends a row $(x\ y)$ to the row $(x\ y)A = (x+4y \quad 2x+5y \quad 3x+6y)$. Written as columns of coordinates in the standard dual bases, this is multiplication by
$$A^{\mathsf{T}} = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}.$$

**Transpose of a product.** $(AB)^{\mathsf{T}} = B^{\mathsf{T}} A^{\mathsf{T}}$, because $(ST)' = T'S'$ (Prop. 47) and the matrix of a dual map is the transpose (Prop. 49).

**Distributions and observables.** For a Markov chain with stochastic matrix $P$ (W6), let $f$ assign a reward $f_i$ to state $i$. Under a distribution $p$ the mean reward is $f^{\mathsf{T}}p = \sum_i f_i p_i$, a linear functional of $p$. After one step the mean reward is
$$f^{\mathsf{T}}(Pp) = (P^{\mathsf{T}}f)^{\mathsf{T}}p, \qquad (P^{\mathsf{T}}f)_j = \sum_i P_{ij} f_i,$$
because $(f^{\mathsf{T}}P)^{\mathsf{T}} = P^{\mathsf{T}}f$ (transpose of a product). So $P$ moves distributions forward, $p \mapsto Pp$, and the dual map moves reward functions, $f \mapsto P^{\mathsf{T}}f$. The entry $(P^{\mathsf{T}}f)_j$ is the average of the rewards $f_i$ weighted by the probabilities $P_{ij}$ of moving from $j$ to $i$: the mean reward one step later when the present state is $j$.

Books that write distributions as rows use $A = P^{\mathsf{T}}$: with initial distribution the row $\lambda^{\mathsf{T}}$, the mean reward after $k$ steps is
$$\lambda^{\mathsf{T}} A^k f = f^{\mathsf{T}} P^k \lambda,$$
the same number in our column convention.

**The question answered.** $\ell \circ T(x, y) = 6x - 3y$. In the standard bases $\mathcal{M}(T) = \begin{pmatrix} 2 & -1 \end{pmatrix}$, so $\mathcal{M}(T')$ is the column $\left[\begin{smallmatrix} 2 \\ -1 \end{smallmatrix}\right]$ (Prop. 49); $\ell$ has coordinate 3, and $3\left[\begin{smallmatrix} 2 \\ -1 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 6 \\ -3 \end{smallmatrix}\right]$ are the coordinates of $6x - 3y$.

**Shortcuts**

- **Matrix of $T'$: transpose.** Reason: Prop. 49. Instance: Ex. 46.

- rank $A^\mathsf{T} = \operatorname{rank} A$, so the span of the rows and the span of the columns have the same dimension. Reason: the dimension count after Prop. 51. Instance: $\left[\begin{smallmatrix}1&2&3\\2&4&6\end{smallmatrix}\right]$ has one independent row and one independent column.

**Traps**

- Orders reverse: $(ST)' = T'S'$ and $(AB)^\mathsf{T} = B^\mathsf{T}A^\mathsf{T}$. With $A = \left[\begin{smallmatrix}0&1\\0&0\end{smallmatrix}\right]$, $B = \left[\begin{smallmatrix}0&0\\1&0\end{smallmatrix}\right]$: $(AB)^\mathsf{T} = \left[\begin{smallmatrix}1&0\\0&0\end{smallmatrix}\right]$, but $A^\mathsf{T}B^\mathsf{T} = \left[\begin{smallmatrix}0&0\\0&1\end{smallmatrix}\right]$.
- $T'$ goes from $W'$ to $V'$, opposite to $T$, and it is not the derivative of $T$: for $T(x,y) = 2x - y$, $T'\ell = 6x - 3y$ (question above).
- $T$ injective does not make $T'$ injective (Midterm 2 review, true/false). $T\colon \mathbb{R} \to \mathbb{R}^2$, $t \mapsto (t, 0)$ is injective, and $T'$ sends the row $(0\ 1)$ to 0. The correct statement is: $T$ injective $\iff$ $T'$ surjective (Prop. 51).

## §2.7 Annihilators (Ch. 10.3)

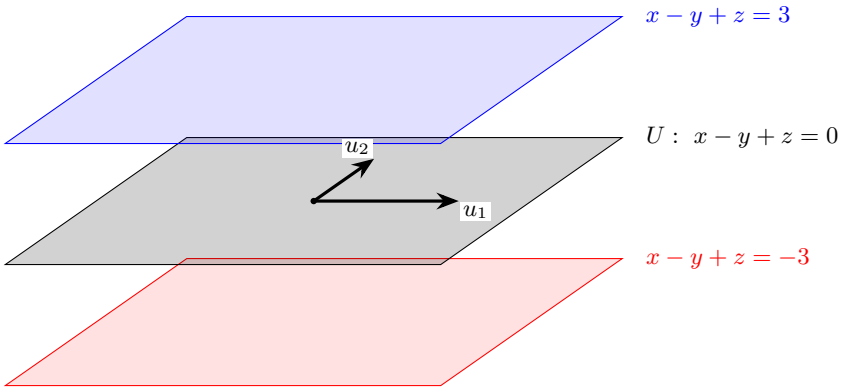
**The question.** Which rows $(a\ b\ c)$ give 0 on both $(1,1,0)$ and $(0,1,1)$? They form a subspace of $(\mathbb{R}^3)'$; what is its dimension, and how is it related to the null space and range of a map?

**Definition** (Def. 29 / LADR 3.121)**.** The **annihilator** of a subspace $U \subseteq V$ is

$$U^0 = \{\varphi \in V' : \varphi(u) = 0 \text{ for all } u \in U\}.$$

- $U^0$ is a subspace of $V'$ (LADR 3.124): the zero functional vanishes on $U$, and if $\varphi, \psi$ vanish on $U$, then $(\lambda\varphi + \psi)(u) = \lambda\varphi(u) + \psi(u) = 0$ for $u \in U$.
- It is enough to check $\varphi$ on a spanning list $u_1, \dots, u_k$ of $U$: then $\varphi(\sum a_i u_i) = \sum a_i \varphi(u_i) = 0$.
- $\{0\}^0 = V'$ and $V^0 = \{0\}$.

**Example 47.** $U = \operatorname{span}\big((1,1,0),(0,1,1)\big)$. A row $(a\ b\ c)$ is in $U^0$ when $a + b = 0$ and $b + c = 0$, that is, $(a\ b\ c) = b(-1\ \ 1\ \ -1)$. So $U^0 = \operatorname{span}\big((-1\ \ 1\ \ -1)\big) = \operatorname{span}\big((1\ \ -1\ \ 1)\big)$.




The plane $U = \operatorname{span}(u_1, u_2)$, $u_1 = (1,1,0)$, $u_2 = (0,1,1)$, and two other level sets $x - y + z = c$ of the functional $(1\ -1\ 1)$, which spans $U^0$. The level set $c = 0$ is $U$ itself: the functional vanishes on $U$.

**Proposition** (LADR 3.125: dimension of the annihilator)
If $V$ is finite-dimensional, $\dim U^0 = \dim V - \dim U$.

**The idea**

Membership in $U^0$ is a condition on the values $\psi(u_1), \dots, \psi(u_k)$. In a dual basis those values are exactly the first $k$ coordinates of $\psi$ (HW5 P4(b)). So choose a basis of $V$ that starts with a basis of $U$: then $\psi \in U^0$ says "the first $k$ coordinates are 0", and the remaining $n-k$ dual vectors form a basis of $U^0$. Midterm 2 Q1 (R3) is this construction with numbers.

*Proof.* 1. Let $u_1, \dots, u_k$ be a basis of $U$; extend to a basis $u_1, \dots, u_k, x_1, \dots, x_r$ of $V$, with dual basis $\varphi_1, \dots, \varphi_k, \chi_1, \dots, \chi_r$.

2. Any $\psi \in V'$ is $\psi = \sum_i \psi(u_i)\varphi_i + \sum_j \psi(x_j)\chi_j$ (coordinates of a functional).

3. $\psi \in U^0$ exactly when $\psi(u_i) = 0$ for all $i$, that is, exactly when $\psi \in \operatorname{span}(\chi_1, \dots, \chi_r)$.

4. $\chi_1, \dots, \chi_r$ is independent (part of a basis), so it is a basis of $U^0$, and $r = \dim V - \dim U$.

□

**Proposition** (LADR 3.127: when the annihilator is everything or nothing)
Let $V$ be finite-dimensional and $U \subseteq V$ a subspace. Then $U^0 = V' \iff U = \{0\}$, and $U^0 = \{0\} \iff U = V$.

*Proof.* 1. By LADR 3.125, $\dim U^0 = \dim V - \dim U$.

2. $U^0 = V'$ exactly when $\dim U^0 = \dim V' = \dim V$ (W1, a subspace of full dimension), that is, when $\dim U = 0$, that is, $U = \{0\}$.

3. $U^0 = \{0\}$ exactly when $\dim U^0 = 0$, that is, $\dim U = \dim V$, that is, $U = V$ (W1).

□

**Theorem** (Prop. 50 / LADR 3.128, 3.130: null space and range of the dual map)
For $T \in \mathcal{L}(V, W)$: (i) $\operatorname{null} T' = (\operatorname{range} T)^0$; (ii) $\operatorname{range} T' = (\operatorname{null} T)^0$.

**The idea**

(i) is a translation. $T'\varphi = 0$ means $\varphi(Tv) = 0$ for all $v$, which says $\varphi$ vanishes on every vector of the form $Tv$, that is, on $\operatorname{range} T$.

(ii) has an easy half and a half that needs a construction. Easy: a functional $\psi \circ T$ vanishes wherever $T$ does. Harder: given $\varphi \in V'$ that vanishes on $\operatorname{null} T$, we must produce $\psi \in W'$ with $\psi \circ T = \varphi$. The equation tells us what $\psi$ has to be on $\operatorname{range} T$: $\psi(Tv) = \varphi(v)$. Is that a valid definition? The same vector of $\operatorname{range} T$ can be $Tv$ and $Tw$; then $v - w \in \operatorname{null} T$, and $\varphi(v) = \varphi(w)$ because $\varphi$ vanishes on $\operatorname{null} T$. So the hypothesis is exactly the well-definedness condition. Outside $\operatorname{range} T$ the equation imposes nothing, so extend $\psi$ in any way.

*Proof.* 1. **(i).** For $\varphi \in W'$:

$$\varphi \in \operatorname{null} T' \iff \varphi \circ T = 0 \iff \varphi(Tv) = 0 \text{ for all } v \in V \iff \varphi \in (\operatorname{range} T)^0.$$

2. **(ii), $\operatorname{range} T' \subseteq (\operatorname{null} T)^0$.** Assume $\varphi \in \operatorname{range} T'$, so $\varphi = T'\psi = \psi \circ T$ for some $\psi \in W'$. For $v \in \operatorname{null} T$: $\varphi(v) = \psi(Tv) = \psi(0) = 0$. So $\varphi \in (\operatorname{null} T)^0$.

3. **(ii), $(\operatorname{null} T)^0 \subseteq \operatorname{range} T'$.** Assume $\varphi$ vanishes on $\operatorname{null} T$. Define $\psi_0 \colon \operatorname{range} T \to \mathbb{F}$ by $\psi_0(Tv) = \varphi(v)$. It is well defined: if $Tv = Tw$, then $v - w \in \operatorname{null} T$, so $\varphi(v) - \varphi(w) = \varphi(v - w) = 0$. It is linear:

$$\psi_0(\lambda Tv + Tw) = \psi_0(T(\lambda v + w)) = \lambda\varphi(v) + \varphi(w) = \lambda\psi_0(Tv) + \psi_0(Tw).$$

4. Extend a basis of $\operatorname{range} T$ to a basis of $W$, and let $\psi \in W'$ equal $\psi_0$ on the basis vectors of $\operatorname{range} T$ and 0 on the added ones. Then $\psi = \psi_0$ on $\operatorname{range} T$. (If $W$ is infinite-dimensional, this uses the cited fact from Prop. 46, step 1.)

5. $T'\psi = \psi \circ T$ sends $v$ to $\psi(Tv) = \psi_0(Tv) = \varphi(v)$. So $T'\psi = \varphi$ and $\varphi \in \operatorname{range} T'$.

□

**Proposition** (Prop. 51 / LADR 3.129, 3.131)

Let $V, W$ be finite-dimensional and $T \in \mathcal{L}(V, W)$. Then $T$ is injective $\iff$ $T'$ is surjective, and $T$ is surjective $\iff$ $T'$ is injective.

**The idea**

Prop. 50 turns statements about $T'$ into statements about annihilators. $T$ injective means $\operatorname{null} T = \{0\}$; its annihilator is all of $V'$, which is $\operatorname{range} T'$. $T$ surjective means $\operatorname{range} T = W$; its annihilator is $\{0\}$, which is $\operatorname{null} T'$. The only thing to check is that taking annihilators loses nothing: $U^0 = V'$ only for $U = \{0\}$, and $U^0 = \{0\}$ only for $U = V$ (LADR 3.127).

*Proof.* 1. The chain for injectivity, using LADR 3.127 in $V$ and Prop. 50(ii):

$$T \text{ injective} \iff \operatorname{null} T = \{0\} \iff (\operatorname{null} T)^0 = V' \iff \operatorname{range} T' = V'.$$

2. The chain for surjectivity, using LADR 3.127 in $W$ and Prop. 50(i):

$$T \text{ surjective} \iff \operatorname{range} T = W \iff (\operatorname{range} T)^0 = \{0\} \iff \operatorname{null} T' = \{0\}.$$

□

The direction "$T$ surjective $\Rightarrow$ $T'$ injective" needs no finite dimension: R2(c).

**Dimensions** (finite dimensions):

$$\begin{aligned} \dim \operatorname{range} T' &= \dim(\operatorname{null} T)^0 = \dim V - \dim \operatorname{null} T = \dim \operatorname{range} T, \\ \dim \operatorname{null} T' &= \dim(\operatorname{range} T)^0 = \dim W - \dim \operatorname{range} T = \dim W - \dim V + \dim \operatorname{null} T. \end{aligned}$$

For a matrix $A$: the range of $A$ is spanned by its columns. By Prop. 49, $A^{\mathsf{T}}$ is the matrix of $T'$, whose range is spanned by the columns of $A^{\mathsf{T}}$, that is, the rows of $A$. So the span of the rows and the span of the columns have the same dimension: $\operatorname{rank} A^{\mathsf{T}} = \operatorname{rank} A$.

**Proposition** (Midterm 2 review: duals of the inclusion and of the quotient map)

Let $U \subseteq V$, $V$ finite-dimensional, $i\colon U \to V$ the inclusion and $\pi\colon V \to V/U$ the quotient map. Then (i) $i'\colon V' \to U'$ is restriction, $\varphi \mapsto \varphi|_U$; it is surjective with null space $U^0$, so $V'/U^0 \cong U'$; (ii) $\pi'\colon (V/U)' \to V'$ is injective with range $U^0$, so $(V/U)' \cong U^0$.

*Proof.* 1. $i'\varphi = \varphi \circ i$ sends $u$ to $\varphi(u)$, so it is $\varphi|_U$.

2. $i$ is injective, so $i'$ is surjective (Prop. 51). $\operatorname{null} i' = (\operatorname{range} i)^0 = U^0$ (Prop. 50(i)). The first isomorphism theorem gives $V'/U^0 \cong U'$.

3. $\pi$ is surjective, so $\pi'$ is injective (Prop. 51). $\operatorname{range} \pi' = (\operatorname{null} \pi)^0 = U^0$ (Prop. 50(ii)). So $\pi'$ is an isomorphism from $(V/U)'$ onto $U^0$.

□

This explains why $\dim U^0 = \dim V - \dim U = \dim(V/U)$.

**The question answered.** The rows are the multiples of $(1\ -1\ 1)$ (Ex. 47), a subspace of dimension $1 = 3 - 2$ (LADR 3.125). If $T\colon \mathbb{R}^2 \to \mathbb{R}^3$ has $\operatorname{range} T = U$ (for example the matrix with columns $(1, 1, 0), (0, 1, 1)$), this subspace is $\operatorname{null} T'$ (Prop. 50(i)).

**Shortcuts**

- **Annihilator in $\mathbb{F}^n$: a linear system.** Solve (row) $\cdot u = 0$ for each spanning vector $u$ of $U$; expect dimension $n - \dim U$. Reason: the bullets after Def. 29 and LADR 3.125. Instance: Ex. 47.
- **Basis of $U^0$ from a dual basis.** Extend a basis $u_1, \dots, u_k$ of $U$ to a basis of $V$; the last $n-k$ dual-basis functionals are a basis of $U^0$. Reason: proof of LADR 3.125. Instance: Midterm 2 Q1 (R3): for $B = (1,1,0), (0,1,1), (1,0,1)$, the third dual vector $\frac{1}{2}(1\ -1\ 1)$ spans $U^0$.
- **Annihilators reverse inclusions.** $U \subseteq W \Rightarrow W^0 \subseteq U^0$, and $(U+W)^0 = U^0 \cap W^0$. Reason: a functional vanishes on $U+W$ exactly when it vanishes on $U$ and on $W$. Condition: none. Instance: in $\mathbb{R}^3$, $(\operatorname{span}(e_1) + \operatorname{span}(e_2))^0$ is the set of rows $(0\ 0\ c)$.
- **Injective and surjective swap under the dual.** $\operatorname{null} T' = (\operatorname{range} T)^0$, $\operatorname{range} T' = (\operatorname{null} T)^0$. Reason: Props. 50, 51. Instance: $T\colon \mathbb{R}^2 \to \mathbb{R}^3$, $T(x,y) = (x,y,0)$ is injective, so $T'$ is onto $(\mathbb{R}^2)'$; $T$ is not surjective, so $T'$ is not injective: $T'$ sends the row $(0\ 0\ 1)$ to 0.

**Traps**

- $U^0 \subseteq V'$, not $V$. For $U = \operatorname{span}((1,1,0))$, $U^0$ is a set of rows (functionals), for example $(1\ -1\ 0)$; it is not the plane of vectors perpendicular to $U$, which is a description that needs the dot product.
- $U^0$ depends on the space $V$ around $U$, not only on $U$: the annihilator of $\operatorname{span}(e_1)$ in $(\mathbb{R}^2)'$ is $\operatorname{span}((0\ 1))$, of dimension 1, while in $(\mathbb{R}^3)'$ it is $\operatorname{span}((0\ 1\ 0), (0\ 0\ 1))$, of dimension 2.

## §3 Problem set

**HW5 P1.** Define $T\colon \mathcal{P}_3(\mathbb{R}) \to \mathbb{R}^2$, $T(p) = (p(0), p(1))$. (a) Determine $\operatorname{null} T$ and prove that $T$ is surjective. (b) Describe explicitly the induced isomorphism $\mathcal{P}_3(\mathbb{R})/\operatorname{null} T \to \mathbb{R}^2$ from the first isomorphism theorem. (c) Find a basis of $\mathcal{P}_3(\mathbb{R})/\operatorname{null} T$.

*Hint:* a polynomial with zeros at 0 and 1 is divisible by $x(x-1)$ (W7). For (c), an isomorphism carries a basis back to a basis.

**HW5 P2.** Let $T\colon V \to W$ be linear and $U$ a subspace of $V$ contained in $\operatorname{null} T$. Prove that there is a unique linear map $\widetilde{T}\colon V/U \to W$ with $\widetilde{T}(v+U) = Tv$ for every $v$. Determine $\operatorname{null} \widetilde{T}$ in terms of $\operatorname{null} T$ and $U$.

*Hint:* well-definedness is the only issue for existence.

**HW5 P3 (LADR 3F, Exercise 1).** Prove that every linear functional $\varphi\colon V \to \mathbb{F}$ is either the zero map or surjective.

*Hint:* scale a vector with $\varphi(v) \neq 0$.

**HW5 P4.** Let $v_1, \dots, v_n$ be a basis of $V$ with dual basis $\varphi_1, \dots, \varphi_n$. (a) Prove $v = \sum_{j=1}^n \varphi_j(v) v_j$ for every $v \in V$. (b) Prove $\psi = \sum_{j=1}^n \psi(v_j)\varphi_j$ for every $\psi \in V'$.

*Hint:* (a) apply $\varphi_j$ to $v = \sum a_k v_k$. (b) evaluate both sides at $v_k$.

**HW5 P5.** Let $T\colon V \to W$ be linear between finite-dimensional spaces with matrix $A$ in bases $B$, $C$. Prove that the matrix of $T'\colon W' \to V'$, $T'(\psi) = \psi \circ T$, in the dual bases is $A^{\mathsf{T}}$.

*Hint:* evaluate $T'(w_j')$ at $v_k$.

**HW5 P6.** Let $U = \operatorname{span}\big((1,1,0), (0,1,1)\big) \subseteq \mathbb{R}^3$. (a) Determine $U^0 \subseteq (\mathbb{R}^3)'$, writing functionals as row vectors. (b) Verify that $\dim U + \dim U^0 = 3$.

*Hint:* a row is in $U^0$ when it gives 0 on both spanning vectors (Ex. 47); for (b) use $\dim U^0 = \dim V - \dim U$.

**HW5 P7 (optional).** Let $j_V\colon V \to V''$, $j_V(v)(\varphi) = \varphi(v)$. Prove that $j_V$ is injective, and an isomorphism when $V$ is finite-dimensional.

*Hint:* for $v \neq 0$ make $v$ a basis vector and take the functional that is 1 on $v$ and 0 on the others (Prop. 46).

**R1 (Practice Midterm 2, Question 1).** Let $V = \mathcal{P}_3(\mathbb{R})$, $E\colon V \to \mathbb{R}^2$, $E(p) = (p(0), p(1))$, $U = \operatorname{null} E$. (a) Prove that $E$ is linear and surjective. (b) Find a basis of $U$. (c) Prove that $\widetilde{E}(p + U) = E(p)$ defines a linear isomorphism $V/U \to \mathbb{R}^2$. (d) Give a basis of $V/U$ and the images of its elements under $\widetilde{E}$.

*Hint:* same map as HW5 P1; (c) is the first isomorphism theorem, checked by hand.

**R2 (Practice Midterm 2, Question 2).** Let $T\colon V \to W$, $S\colon W \to Z$ be linear maps between finite-dimensional real spaces, with dual maps $T^*\colon W^* \to V^*$, $S^*\colon Z^* \to W^*$. (a) Give the definition of $T^*$ and prove it is linear. (b) Prove $(S \circ T)^* = T^* \circ S^*$. (c) Prove that if $T$ is surjective then $T^*$ is injective. (d) For $T\colon \mathbb{R}^2 \to \mathbb{R}$, $T(x, y) = 2x - y$ and $\ell(t) = 3t$, compute $T^*\ell$.

*Hint:* everything follows from $(T^*\varphi)(v) = \varphi(Tv)$; for (c), $T^*\varphi = 0$ means $\varphi$ vanishes on $\operatorname{range} T = W$.

**R3 (Midterm 2, Question 1).** Let $V = \mathbb{R}^3$ and $U = \operatorname{span}\big((1, 1, 0), (0, 1, 1)\big)$. (a) What is meant by $U^0$, the annihilator of $U$ in $V'$? (b) Find a basis of $U^0$. (c) Extend the two vectors to a basis $B$ of $V$ and find its dual basis. (d) For the basis $B$ chosen in (c), explain why the third element of its dual basis spans $U^0$.

*Hint:* (c) put $B$ as the columns of $M$; the dual basis is the rows of $M^{-1}$. (d) the third dual vector vanishes on the first two basis vectors, which span $U$; then count dimensions with LADR 3.125.

**R4 (Midterm 2 review, Problem 1).** Let $V = \operatorname{span}(1, \sin x, \sin 2x)$, real functions on $[-\pi, \pi]$. (a) Show that $\delta_0(f) = f(0)$, $\delta_{\pi/2}(f) = f(\pi/2)$, $\delta_0'(f) = f'(0)$ form a basis of $V'$. (b) Write the dual basis $\varphi_1, \varphi_2, \varphi_3$ of $1, \sin x, \sin 2x$ as combinations of $\delta_0, \delta_{\pi/2}, \delta_0'$. (c) For the functional $\varphi(f) = \frac{1}{\pi}\int_0^\pi f(x)\,dx$, write $\varphi$ in the basis $\varphi_1, \varphi_2, \varphi_3$ and then in $\delta_0, \delta_{\pi/2}, \delta_0'$.

*Hint:* record the values $\delta_i(b_j)$ in a $3 \times 3$ matrix; three functionals in a 3-dimensional $V'$ are a basis when that matrix is invertible, and its inverse gives the dual basis. For (c) use HW5 P4(b).

**R5 (Midterm 2 review, Problem 2).** Let $U \subseteq V$ with $V$ finite-dimensional, $i\colon U \to V$ the inclusion and $\pi\colon V \to V/U$ the quotient map. (a) Describe $i'$ and show it is surjective with null space $U^0$. (b) Show $\pi'$ is injective with range $U^0$. (c) Conclude $V'/U^0 \cong U'$ and $(V/U)' \cong U^0$.

*Hint:* Prop. 50 and Prop. 51 applied to $i$ and $\pi$; then the first isomorphism theorem.

**R6 (Midterm 2 review, true or false).** Decide, with a proof or a counterexample: (a) if $T \in \mathcal{L}(V, W)$ is injective, then $T'$ is injective; (b) if $T$ is surjective, then $T'$ is injective; (c) for finite-dimensional $V$ and a subspace $U$, $\dim U^0 = \dim(V/U)$.

*Hint:* (a) try $T\colon \mathbb{R} \to \mathbb{R}^2$, $t \mapsto (t, 0)$; (b) Prop. 51 or R2(c); (c) LADR 3.125 and Prop. 40.

**E1 (dual basis computation).** Find the dual basis of the basis $v_1 = (2, 1)$, $v_2 = (1, 1)$ of $\mathbb{R}^2$, as row vectors. Then write the functional $\psi(x, y) = 3x + y$ in that dual basis.

*Hint:* the dual basis is the rows of $M^{-1}$ for $M$ with columns $v_1, v_2$; the coordinates of $\psi$ are $\psi(v_1), \psi(v_2)$.